



Food Delivery Time Prediction

Analyzing delivery performance to identify delay patterns and predict estimated delivery time

Project Overview

Goal

Identify factors affecting delivery time and predict ETA

Scope

Restaurants, locations, vehicle types, time periods

Impact

Improve operational efficiency and customer satisfaction



Dataset Summary

50K

Rows

Delivery transactions

24

Columns

Features & metrics

1

Target

Time to Deliver

Order Details

ID, restaurant, date/time, food category

Delivery Details

Distance, pickup time, vehicle type

External Factors

Traffic, weather, location

Customer & Performance

Ratings, prep time, delay status

Exploratory Data Analysis

```
[2]:
```

	Order_ID	Order_Date	Order_Hour	Day_of_Week	Is_Weekend	Is_Festival	Weather	Pickup_Zone	Dropoff_Zone	Vehicle_Type	...	Order_Items	Restaurant_Load
0	ORD000001	2025-10-28	8	Tuesday	0	0	Clear	CBD	Commercial	Scooter	...	1	Medium
1	ORD000002	2025-09-26	12	Friday	0	0	Rain	CBD	CBD	Scooter	...	3	Medium
2	ORD000003	2025-01-26	18	Sunday	1	0	Clear	Residential	Residential	Scooter	...	1	Low
3	ORD000004	2025-11-19	12	Wednesday	0	0	Cloudy	Residential	Industrial	Bike	...	3	Medium
4	ORD000005	2025-08-04	18	Monday	0	0	Rain	Industrial	Residential	Bike	...	1	Medium

5 rows × 24 columns

Data Loading & Initial Exploration

Imported dataset via `pandas`, used `df.info()` and `.describe()` for structure and summary statistics.

Data Preprocessing

→ Missing Data

Null values checked and imputed

→ Standardization

Columns renamed to snake_case

→ Feature Engineering

vehicle_type, cuisine_type,
prep_time, ride_ratings

→ Consistency Check

Duplicates, invalid times verified

→ Database Integration

Loaded into PostgreSQL for SQL &
Power BI

```
<class 'pandas.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 24 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Order_ID                             50000 non-null  str
1   Order_Date                           50000 non-null  str
2   Order_Hour                           50000 non-null  int64
3   Day_of_Week                          50000 non-null  str
4   Is_Weekend                          50000 non-null  int64
5   Is_Festival                          50000 non-null  int64
6   Weather                             50000 non-null  str
7   Pickup_Zone                         50000 non-null  str
8   Dropoff_Zone                        50000 non-null  str
9   Vehicle_Type                        50000 non-null  str
10  Rider_Experience_Years               50000 non-null  float64
11  Rider_Rating                        50000 non-null  float64
12  Restaurant_Rating                   50000 non-null  float64
13  Cuisine_Type                        50000 non-null  str
14  Order_Items                         50000 non-null  int64
15  Restaurant_Load                     50000 non-null  str
16  Preparation_Time_Min                50000 non-null  int64
17  Road_Distance_km                   50000 non-null  float64
18  Delivery_Distance_Category          50000 non-null  str
19  Traffic_Level                       50000 non-null  str
20  Number_of_Signals                   50000 non-null  int64
21  Average_Speed_kmph                  50000 non-null  float64
22  Delivery_Priority                   50000 non-null  str
23  Time_taken_min                      50000 non-null  int64
dtypes: float64(5), int64(7), str(12)
memory usage: 13.1 MB
```

SQL Analysis — Orders & Delivery Time

	pickup_zone text 	sum numeric 
1	Suburban	17386
2	Residential	41571
3	CBD	29085
4	Industrial	19501
5	Commercial	25094

Orders per Restaurant

Order volume by restaurant

Avg Delivery Time by Vehicle

Performance across vehicle types

Avg Delivery Time by Weather

Impact of weather conditions

Orders Under 30 Minutes

% of fast deliveries

SQL Analysis — Weather, Zones & Peak Hours

	weather text	round numeric
1	Storm	114.45
2	Rain	98.86
3	Clear	77.30
4	Cloudy	79.02
5	Fog	84.15

Top 5 Slowest Restaurants

Highest average delivery time

Top 5 Pickup Zones in Rain

Most orders during rainy weather

Fastest & Slowest by Zone

Top 5 per pickup zone

Peak Ordering Hour

Order volume by hour

SQL Analysis — Delays & Traffic

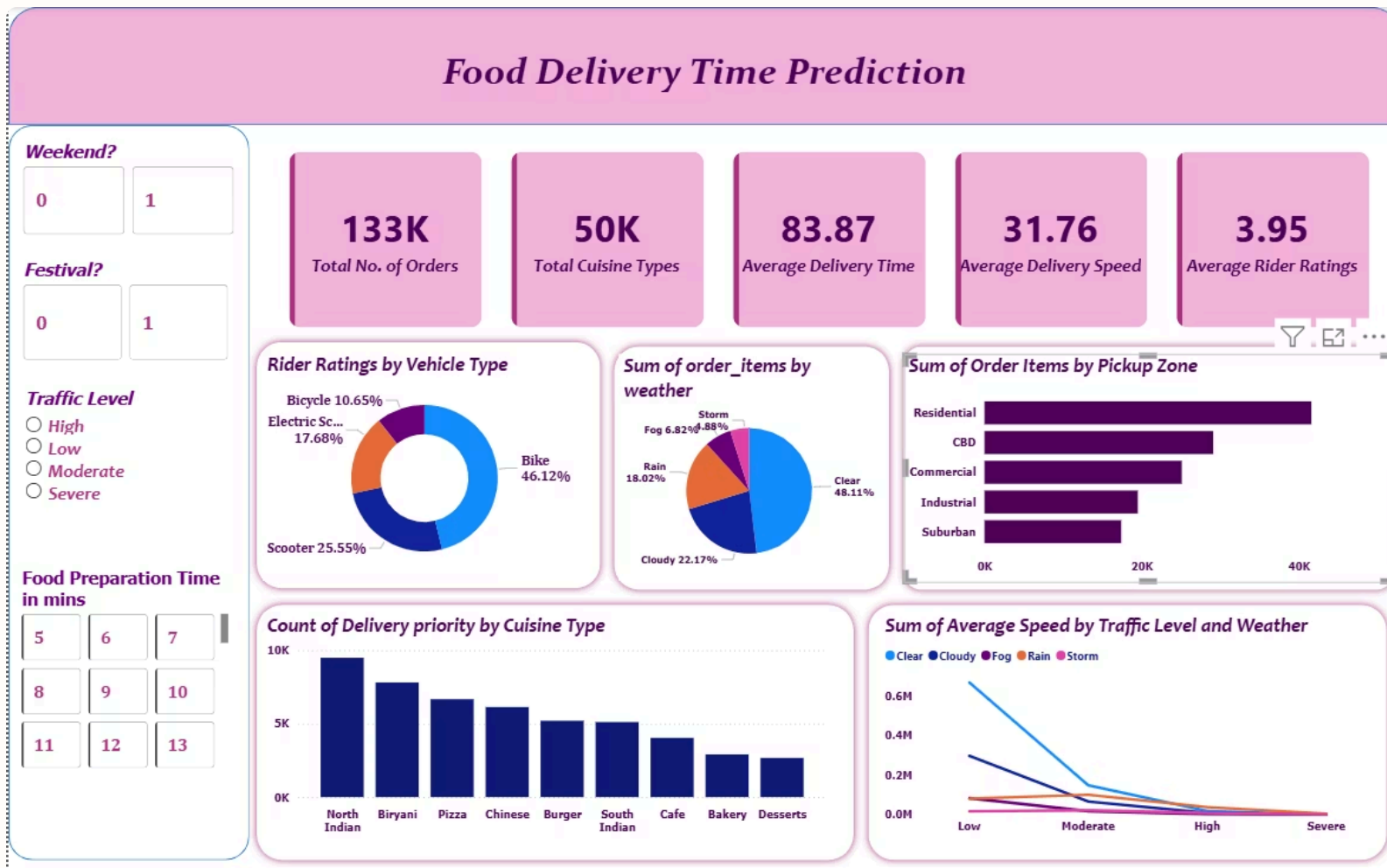
	pickup_zone text 	delayed_orders bigint 
1	Suburban	5741
2	Residential	13800
3	CBD	10408
4	Industrial	6105
5	Commercial	8469

Delayed Orders (>45 min)

Count of significantly delayed deliveries

Orders by Traffic Level

Distribution across traffic conditions



Power BI Dashboard

Interactive visual reporting layer connecting SQL analysis to business stakeholders

Business Recommendations



Optimize Operations

Allocate partners by volume, location, peak hours



Improve Restaurants

Monitor and coach high-delay locations



Peak-Hour Planning

Increase partner availability during high demand



Reduce Delays

Address traffic, weather, distance, prep time



Vehicle Optimization

Match vehicle type to distance and traffic



Customer Experience

Accurate ETAs and performance insights